Sanghyun Kim

Curriculum Vitae

Postdoctoral Researcher

Center for KVN, Division of Observational Infrastructure Research,

Korea Astronomy and Space Science Institute (KASI)

776 Daedeok-daero, Yuseong-gu, Daejeon, 34055 Republic of Korea

sanghkim.astro@gmail.com +82-10-3399-7480 sanghkimastro.github.io ORCID

Personal Data

- Birth: 16 Dec 1993, in Republic of Korea
- Nationality: Republic of Korea
- Language: Korean (native); English - read, spoken, and written

Education

Ph.D. **University of Science and Technology** Mar 2018 – Feb 2025
Korea Astronomy and Space Science Institute
Republic of Korea

- M.Sc./Ph.D. Integrative Course
- Major: Astronomy and Space Science
- Thesis: *Multi-Wavelength and Multi-Messenger Study of the Blazars CTA 102 and PKS 0735+178*
- Supervisor: Prof. Sang-Sung Lee

B.Sc. **Kyungpook National University** Mar 2012 – Feb 2018
Republic of Korea

- Major: Astronomy and Atmospheric Sciences
- Minor: Physics
- Thesis: *Mass of Black Hole in Quasars*
- Supervisor: Prof. Myeong-Gu Park

Employment

Korea Astronomy and Space Science Institute Republic of Korea
Postdoctoral Researcher, Center for KVN, Observational Infra Research Division Mar 2025 – present

Korea Astronomy and Space Science Institute Republic of Korea
Student Researcher, Radio Astronomy Division Mar 2018 – Feb 2025

- M.Sc. & Ph.D. (integrative) Thesis Research (full-time)
- Supervisor: Prof. Dr. Sang-Sung Lee

Grants and Awards

Invitation for Collaboration 1–9 Nov 2025
Host Institution: INAF-IRA, Bologna, Italy
Host Researcher: Dr. Marcello Giroletti

UST Outstanding Research Assistant (RA) Dec 2024
The 2024 fall semester
University of Science and Technology, Republic of Korea

UST Overseas Training Program Nov 2023 – Feb 2024
Host Institution: Julius-Maximilians-Universität Würzburg, Germany
Host Professor: Prof. Matthias Kadler

Best Student Paper Award Sep 2023
The 49th Founding Anniversary
Korea Astronomy and Space Science Institute, Republic of Korea

Research Interests

- VLBI correlation
- VLBI data calibration and pipeline development
- Multi-wavelength and multi-messenger astrophysics of AGN jets
- VLBI jet kinematics
- Light curve and variability analysis of AGNs
- Multi-wavelength polarimetry of AGN jets

Collaboration and Working Group Membership

- **K**orean **V**LBI **N**etwork (**KVN**) Array Operation Center, since 2025
- **E**ast **A**sia **V**LBI **N**etwork (**EAVN**) Array Operation Center, since 2025
- EAVN AGN Science Working Group, since 2025
- Fermi Large Area Telescope Collaboration, since 2025
- **T**eV **E**ffelsberg **L**ong-term **A**gn **MON**itoring (**TELAMON**), since 2023
- **I**nterferometric **MO**nitoring of **GA**mma-ray **B**right **A**GNs (**iMOGABA**), since 2018
- **MO**nitoring of **GA**mma-ray **B**right **A**GNs (**MOGABA**), since 2018

Conferences and Seminars

Contributed Talks

- ***Multimessenger Study on Blazars as Promising Neutrino Emitters with VLBI: the Case of PKS 0735+178 and PKS 0446+11***
  16th East Asian VLBI Workshop 2025
  PyeongChang, Republic of Korea, 17–20 November 2025
- ***Probing the origin of high-energy emission in blazar jets using high-resolution VLBI***
  Special Colloquium at the INAF-IRA
  Bologna, Italy, 3 November 2025
- ***Multimessenger study of neutrino-candidate blazar PKS 0735+178 using Korean VLBI Network multi-band observations***
  Towards high-performance mm-VLBI science operations with multi-band receivers
  Bologna, Italy, 28–31 October 2025
- ***Multimessenger study of the blazar PKS 0735+178 as a potential neutrino source using the Korean VLBI Network***
  Extragalactic jets in Crete 2025
  Heraklion, Greece, 25–29 August 2025
- ***Multi-Wavelength and Multi-Messenger Study of the Blazars CTA 102 and PKS 0735+178*** - PhD Thesis
  2025 111th Korean Astronomical Society Spring Meeting
  Gyeongju, Republic of Korea, 9–11 April 2025
- ***Gamma-ray flares from the jet of the blazar CTA 102 in 2016–2018***
  Radio Seminar, Korea Astronomy and Space Science Institute
  Daejeon, Republic of Korea, 25 Feb 2025
- ***Multiwavelength Variability and VLBI Studies of the Blazars CTA 102 and PKS 0735+178***
  Galaxy Group Meeting at the Max Planck Institute for Astrophysics
  Garching, Germany, 31 Jan 2024
- ***Multiwavelength Variability and VLBI Studies of the Blazars CTA 102 and PKS 0735+178***
  Special Colloquium at the Max Planck Institute for Radio Astronomy
  Bonn, Germany, 24 Jan 2024
- ***Magnetic field strengths of the synchrotron self-absorption region in the jet of CTA 102 during radio flares***
  2023 Radio summer school & Radio telescope users' meeting
  Siheung, Republic of Korea, 21–23 Aug 2023
- ***External Compton Origin of the Gamma-ray Flares from the Jet of the Blazar CTA 102 in 2016–2018***
  2023 107th Korean Astronomical Society Spring Meeting
  Jeonju, Republic of Korea, 12–14 Apr 2023
- ***Correlations of the multiwavelength emission in the blazar CTA 102 during 2016–2018***
  15th EVN Symposium & Users Meeting: Providing the sharpest View of the Universe
  Cork, Ireland, 11–15 Jul 2022
- ***Magnetic Field Strengths of Flaring Region in the Jet of CTA 102***
  2021 103th Korean Astronomical Society Spring Meeting
  virtual, Republic of Korea, 15–16 Apr 2021

- ***Magnetic Field Strengths of Flaring Region in the Jet of CTA 102***
  13th East Asian VLBI Workshop 2021
  virtual, Chiang Mai, Thailand, 2–5 Mar 2021

Posters

- ***Probing the High-Energy Neutrino Connection in Blazar PKS 0446+11 with Multi-Band Spectrum and Polarization***
  2025 112th Korean Astronomical Society Fall Meeting
  Jeju, Republic of Korea, 15–17 Oct 2025
- ***Interferometric Monitoring of a Potential Neutrino-Emitting Blazar PKS 0735+178: a Connection between Neutrino Events and Radio Flares?***
  16th EVN Symposium & Users Meeting
  Bonn, Germany, 2–6 Sep 2024
- ***Interferometric monitoring of a blazar PKS 0735+178: a connection between neutrino and radio flares?***
  2023 108th Korean Astronomical Society Fall Meeting
  Jeju, Republic of Korea, 18–20 Oct 2023
- ***Correlations of the multiwavelength emission in the blazar CTA 102 during 2016–2018***
  International Astronomical Union General Assembly (IAUGA) 2022
  Busan, Republic of Korea, 2–11 Aug 2022
- ***Correlations of the multiwavelength emission in the blazar CTA 102 during 2016–2018***
  2022 105th Korean Astronomical Society Spring Meeting
  Busan, Republic of Korea, 13–15 Apr 2022
- ***Magnetic Field Strength of Flaring Region in the Jet of CTA 102***
  European Astronomical Society Annual Meeting 2021
  virtual, 28 Jun – 28 Jul, 2021
- ***Interferometric Monitoring of Gamma-ray Bright AGNs (iMOGABA): A multi-frequency study of CTA 102 during the most prominent gamma-ray flare activity in 2016–2017***
  12th East Asian VLBI Workshop 2019
  Ibaraki, Japan, 23–27 Sep 2019
- ***Identification Of Jet Components Of CTA 102 On Milliarcsecond Scales Using The iMOGABA Program***
  2019 100th Korean Astronomical Society Spring Meeting
  Busan, Republic of Korea, 10–12 Apr 2019
- ***Interferometric Monitoring of Gamma-ray AGNs: CTA 102***
  2018 99th Korean Astronomical Society Fall Meeting
  Cheongsong-gun, Republic of Korea, 10–12 Oct 2018

Successful Observing Proposals

List of Observational Proposals as a **PI**

- **EAVN** Semester 2026A Proposal for 60 hours
  : *High-Resolution VLBI Probes of Parsec-Scale Jets in TeV Blazars*
- **EAVN** Semester 2025B Proposal for 60 hours
  : *High-Resolution VLBI Probes of Parsec-Scale Jets in TeV Blazars*
- **KVN** Semester 2023B Re-observations of the Semester 2022B Proposal for 30 hours
  : *Monthly Monitoring of the Neutrino Emitting Blazar 0735+178*
- **KVN** Semester 2023A Director's Discretionary Target (DDT) Proposal for 5 hours
  : *Flux Measurements of the Neutrino-emitting Blazar 0735+178*
- **KVN** Semester 2022B Proposal for 60 hours
  : *Monthly Monitoring of the Neutrino Emitting Blazar 0735+178*
- **KVN** Semester 2022A Target of Opportunity (ToO) Proposal for 20 hours
  : *KVN Follow-up of Neutrino-emitting Blazar 0735+178*
- **KVN** Semester 2020A Proposal for 12 hours
  : *Multi-frequency study of fast jets in AGNs at 22–129 GHz*
- **KaVA** Semester 2019A Proposal for 20 hours
  : *KaVA VLBI Monthly Observation of 3C 120 for jet kinematics*

List of Observational Proposals as a **Collaborator**

- **Effelsberg 100-m Telescope** KSP Proposal for 3 years (PI: Prof. Matthias Kadler)
  : *TELAMON: Monitoring of AGN Jets with Very-High-Energy Astroparticle Emissions*
  * Note. This co-authorship is granted through membership in the TELAMON program.
- **KVN** Semester 2020A–2026A Proposals (PI: Dr. Sincheol Kang)
  : *Monitoring of Gamma-ray Bright AGNs (MOGABA)*
  * Note. This project has been conducted in collaboration with the *POLAMI* and *IXPE* programs.
- **KVN** Semester 2025B Proposal for 84 hours (PI: Prof. Jae-Young Kim)
  : *Probing neutrino-AGN connection by millimeter-wave monitoring*
- **KVN** Semester 2025B Proposal for 12 hours (PI: Hyeon-Woo Jeong)
  : *Test on polarization calibration through D-term transfer*
- **GMVA** Semester 2025B–2026A Proposal for 72 hours (PI: Florian Eppel)
  : *mm-VLBI monitoring of TeV-emitting AGN*
- **GMVA** Semester 2025B Proposal for 10 hours (PI: Dr. Luca Ricci)
  : *First 7 mm image of Cen A: investigating the jet polarization and geometry*
- **KVN** Semester 2024B Proposal for 60 hours (PI: Prof. Jae-Young Kim)
  : *Probing the physical origin of TeV/PeV neutrinos from AGNs by Faraday rotation measure*
- **GMVA** Semester 2024B Proposal for 10 hours (PI: Dr. Luca Ricci)
  : *First 7 mm image of Cen A: investigating the jet polarization and geometry*
- **GMVA** Semester 2024B Proposal for 18 hours (PI: Florian Roesch)
  : *Probing Characteristic Signatures of Neutrino Production in Blazars*
- **GMVA** Semester 2024A Proposal for 16 hours (PI: Florian Roesch)
  : *Probing Characteristic Signatures of Neutrino Production in Blazars at 3 mm*
- **KVN** Semester 2023B ToO Proposal for 21 hours (PI: Dr. Sang-Sung Lee)

: *PKS 0446+11 ToO*

Observation Experiences

- **Korean VLBI Network**, Republic of Korea
  - VLBI operation support as a postdoctoral researcher at the Center for KVN in KASI, Nov 2025 – present
  - VLBI operation support as a PhD student, Feb 2019 – Feb 2020
  - Single-dish polarization as part of the ***MOGABA*** program
- **Mopra 22-m radio telescope**, Australia
  - VLBI observations connected with the KVN at 22 GHz
- **Effelsberg 100-m radio telescope**, Germany
  - Single-dish continuum and polarization observations as part of the ***TELAMON*** program

Computer Skills

- **Programming Languages**: Python, Bash, LaTeX (Overleaf)
- **Astronomical Software**: astropy, ehtim (eht-imaging), SCHED, GILDAS-CLASS, AIPS, DIFMAP, ParselTongue
- **Software**: emcee, macOS, Linux, Windows, Microsoft Office

Academic Service

- **LOC Member** of Event Horizon Telescope Collaboration Meeting 2026
- **LOC Member** of 16th East Asian VLBI Workshop 2025, PyeongChang, Korea
- **LOC Member** of Opening Ceremony of the KVN Pyeongchang Radio Telescope, Seoul National University Pyeongchang Campus, Korea
- **LOC Member** of The 10th Anniversary of the Korean VLBI Network (KVN), Korea Astronomy and Space Science Institute in Daejeon, Korea
- **LOC Member** of 2019 Radio Summer School & Radio Telescope User's Meeting, Seoul National University in Pyeongchang, Korea
- **LOC Member** of East Asian VLBI Workshop 2018, PyeongChang, Korea

Other Experience

Military Service: *Republic of Korea Army* Sep 2013 – Jun 2015

Academic References

- **Prof. Sang-Sung Lee**
  sslee@kasi.re.kr
  Center for Star Formation and Evolution
  Fundamental Astronomy and Space Science Division
  Korea Astronomy and Space Science Institute

  Department of Astronomy and Space Science
  University of Science and Technology, Korea
- **Prof. Jae-Young Kim**
  jaeyoungkim@unist.ac.kr
  Department of Physics
  Ulsan National Institute of Science and Technology
- **Dr. Juan Carlos Algaba**
  algaba@um.edu.my
  Department of Physics
  Faculty of Science
  Universiti Malaya
- **Dr. Bindu Rani**
  bindu.rani@nasa.gov
  NASA Goddard Space Flight Center

  Center for Space Science and Technology
  University of Maryland, Baltimore County